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An investigation of adoption intention of halal traceability system among food SMEs

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Abstract

Purpose – The incredulity among Muslim consumers due to fake and doubtful halal logos has led to some querying the halal compliance and halal integrity among food small and medium enterprises (SMEs). By using the traceability systems consumers may track and trace the movement of food products available in the market. The purpose of this paper is to investigate factors that influence food SMEs' intention to adopt a halal traceability system.

Design/methodology/approach – A structured questionnaire survey was developed and administered to a systematic random sampling of 260 food SMEs. The data were analyzed using descriptive analysis, Chi-square analysis, Pearson correlation analysis and logistic regression analysis.

Findings – The results revealed a strong correlation between the environmental aspect (EA) and perceived usefulness (PU) of a halal traceability system. Sales turnover, PU, perceived ease of use, technological aspect, organizational aspect and EA are the factors that influenced food SMEs' intention to adopt a halal traceability system.

Research limitations/implications – The context of this study is confined to the SMEs in the food industry in Peninsular Malaysia, thereby limiting the generalizability of the findings to other industries.

Practical implications – This study shows a halal traceability system facilitates food SMEs in enhancing their business and provides tremendous potential to further improve the halal industry in Malaysia.

Originality/value – The traceability system that is perceived to be easy and useful are the most influential factors toward the adoption of technology among food SMEs. Thus, this study confirms the growing importance of the halal traceability system in the food industry.

Keywords Food SMEs, Adoption, Intention, Halal, Traceability system

Paper type Research paper

1. Introduction

The halal food industry had captivated the attention of entrepreneurs as the industry showed impressive development over time. Nowadays, consumers have become well-

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informed and health-conscious consumers as the demand for healthy and safe food hike up. Hence, the Malaysian halal food industry provides a great platform for Malaysian producers and manufacturers, specifically small and medium enterprises (SMEs) to initiate a business. Halal food production entails the Shariah food compliance that emphasizes the hygiene and safety of the products that are produced by producers and manufacturers (Abdul Talib *et al.*, 2014; Ali and Suleiman, 2016; Shafii and Khadijah, 2011). The involvement of SMEs in the halal food industry would satisfy the demand for halal products in the country, and therefore, afford Muslim consumers in Malaysia access to the products, especially food without being skeptical. Moreover, the religious obligation has influenced Muslim consumers to be vigilant in appraising their dietary consumption; hence, respecting the halalness of the products as the issue is very sensitive. However, the violation of halal food products has caused an immense concern among Muslim consumers where the occurrence would be a threat for food producers and manufacturers (Mohayidin and Kamarulzaman, 2014).

Skepticism among Muslim consumers in some way was triggered due to the fake and doubtful halal logos, which led to some querying the halal compliance and halal integrity among halal industry players (Ali *et al.*, 2013; Mohamad and Hassan, 2011; Rajendran and Kamarulzaman, 2019), as well as halal traceability system in the Malaysian markets (Rashid *et al.*, 2018; Mohamad *et al.*, 2020; Zailani *et al.*, 2010). To prevent the recurrence of the issues, establishing an effective traceability system in Malaysia is, therefore, extremely important (Rashid *et al.*, 2018; Mohamed *et al.*, 2016; Poniman *et al.*, 2015; Shafii and Khadijah, 2011; Zailani *et al.*, 2010). Halal transparency along the food supply chain could be enhanced by a halal traceability system that effectively encourages trust in consumers toward halal food products and food safety (Nawi *et al.*, 2018). Halal supply chain management ensures the continuous status of halal on food products along the companies' supply chain. A fully integrated supply chain is imperative to ensure both inbound and outbound logistics are in line with the halal traceability system, as well as sustaining halal status together with product information across the supply chain (Abdul Majid *et al.*, 2019; Ahmad Tarmizi *et al.*, 2014; Usman *et al.*, 2018).

The Malaysian food industry must establish a model of the local halal food supply chain to ensure and enhance the effectiveness of a system within the country (Abdul Rashid *et al.*, 2018; Ali *et al.*, 2013; Mohamed *et al.*, 2016). Besides, the framework of the halal traceability system is a tool that benefits all parties across the food supply chain for both who are directly and indirectly involved in the manufacturing of halal food (Abdul Rashid *et al.*, 2018; Poniman *et al.*, 2015). Halal traceability system along the food supply chain is crucial as it can mitigate the occurrence of halal issues in the food industry, namely, product contamination, fake halal logos and the lack of trust among consumers regarding halal food products (Abdul Rashid *et al.*, 2018; Abd Rahman *et al.*, 2017). Hence, the development of halal traceability and halal tracking is gradually occurring. Besides, it is crucial and emerged as an important element in the global halal market (Zailani *et al.*, 2010). All players across the halal supply chain could conveniently trace and track any necessary processes in an integrated traceability system (Abdul Rashid *et al.*, 2018; Rahman and Abdul, 2017). In turn, there must be efficient handling of documents and communication between the players to ensure that the product information could be traced and tracked (Usman *et al.*, 2018). Reinforcement for several components, namely, awareness, understanding, action and transparency, are required and that these crucial elements must be strengthened as the foundation for a halal traceability system across the supply chain (Rashid *et al.*, 2018; Samsi *et al.*, 2011). The adoption of the halal traceability system of a company is signified by the concurrent execution of a full-scale traceability system and the compliance of shariah

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requirements (Shafii and Khadijah, 2011). Radio Frequency Identification (RFID) is typically one of the various devices required for a full-fledged halal traceability system (Zailani *et al.*, 2010).

Halal assurance management system (HAS) adheres to the halal requirements and allowing the availability of traceability throughout the food supply chain. The HAS was developed by the Department of Islamic Development Malaysia (JAKIM) (2011) to ensure companies adopt a system to improve their efficiency in controlling the authenticity and purity of halal. Such action was taken to enhance the traceability system for business entities in the Malaysian market. However, therein lies a problem concerning the effectiveness of the guideline toward food SMEs. Both halal-related agencies, SME Corporation Malaysia and Halal Development Corporation Berhad (HDC) indicated that the inconsistency composition of SMEs for the respective categories wherein small and micro sizes represented the most of SMEs among the three sizes classified under SMEs in Malaysia. The problem, however, arises whenever these data were not aligned with the requirements of HAS implementation as stated in the manual of halal certification procedure Malaysia (Third Revision) (JAKIM, 2015). The manual highlighted those small and micro-size companies are not required and bound to HAS implementation, whereas there is a necessity for these two categories, the multinational companies (MNCs) and medium-size companies to implement HAS. This indicates that only small representatives of SMEs are exposed and familiar with HAS, which is a systematic system that could enhance their business efficiency. This scenario would also lead micro and small size SMEs to not prioritize the implementation of HAS. Besides, a low workforce, as well as low skilled labor, would discourage micro and small size SMEs from implementing HAS as they will find it not necessary.

The adoption of a traceability system based on a web system is more efficient and effective compared to a paper-based system (Gunnar and Fremme, 2007; Neumuller and Grunbacher, 2006). However, many companies that adopt traceability systems commonly used manual traceability tools and methods which is inefficient compared to semi-automated and automated approaches (Neumuller and Grunbacher, 2006; Ramdani *et al.*, 2013). This accords with the revelation made by SME Corporation Malaysia (2018) that the adoption of advanced systems by most of the SMEs in Malaysia that has a similar level of sophistication as the traceability system is still low. The systems include intranet, extranet, e-commerce marketplace and customer relationship management, even though smartphones, computers and the internet have been used widely for their business operations. The strong refusal to learn information technology that afflicts SMEs in Malaysia can be attributed to the concerns on the forthcoming challenges that need to be faced later (Thurasamy *et al.*, 2009). The concerns arise from the constraints faced by SMEs as they have an unconvincing technological base, limited use of information technologies and poor human resources (Ahmad Tarmizi *et al.*, 2020; Chin *et al.*, 2012). Furthermore, the Organisation for Economic Co-operation and Development (OECD) (2019) attributed several reasons that cause the lagged in digitalization among SMEs, namely, less preparation in data protection to face cyber security threats, the struggle in integrating various types of innovation, unskilled in managing digital transformation, less engagement of employees with ICT training, as well as limited skilled employees, particularly in communication and management. These constraints have indeed influenced SMEs' intention to adopt new technologies in their business.

On the other end, consumers are primarily concerned about the violation of halal food products by producers and manufacturers across the supply chain. Muslim consumers, who form the majority of the population in Malaysia, tend to be more delicate and responsive in

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their selection of food products for consumption. Therefore, to retain and regain the customers' trust, manufacturers and producers, particularly SMEs, must seriously contemplate this situation. The government had taken an approach to introduce Malaysian SMEs to the global spotlight by exposing them to several advanced technologies. The halal traceability system is perhaps the most important technology that requires urgent introduction to ensure transparency across the halal food supply chain (Rashid *et al.*, 2018; Mohamad *et al.*, 2020). Furthermore, the adoption of advanced technology systems by SMEs is still low (Ahmad Tarmizi *et al.*, 2020; Ramdani *et al.*, 2013). Interestingly, there must have been a gap somewhere that led to this situation as the use of computers, smartphones among Muslim consumers and the internet among SMEs is exceptionally high (SME Corporation Malaysia, 2018). The apprehension is on the existence of barriers that led to the low adoption of these systems which indirectly influencing their intention to adopt the halal traceability system. Hence, in light of the above discussions, we established four specific objectives for this study: -

(1) to identify the intention level of food SMEs to adopt a halal traceability system;

(2) to determine an association between firmographic profiles of food SMEs and their intention to adopt a halal traceability system;

(3) to determine the relationship between technological aspect (TA), organizational aspect (OA) and environmental aspect (EA) toward perceived usefulness (PU) and perceived ease of use (PEOU); and

(4) to investigate the factors that influence food SMEs' intention to adopt a halal traceability system.

The next section of this paper presents the various determinants of the intention to adopt a halal traceability system among food SMEs. Accordingly, the relevant hypotheses and literature that supports this research are described. In the following sections, the research methodology and findings are presented. Finally, we present the implications, limitations and conclusions of the research.

2. Literature review

2.1 Food traceability system

Traceability system is defined as "The ability to follow the movement of products through specified stage(s) of production, processing and distribution" (International Organization for Standardization (ISO), 2007). As ISO definition is more generic, product-based traceability is often used in most of the literature as researchers tried to define traceability systems respecting their research works (Khan *et al.*, 2018). Traceability system connects all players in the supply chain, therefore, it is crucial to improve the level of food security and food safety (Cao *et al.*, 2017). Besides, a traceability system could improve the performance of companies by minimizing the risk of a product recall (Dabbene *et al.*, 2014). Once consumers perceived the value of the food traceability system, they intend to purchase traceable food (Yuan *et al.*, 2020). Zhang and Zou (2017) claimed that the food traceability system along the supply chain was placed under scrutiny because of the recurrence of the food crisis. Consumers can be protected from food hazard and product adulteration once the documentation regarding food products are properly handled (Haleem *et al.*, 2019). Besides, there are several barriers for a firm to adopt a traceability system, namely, lack of expertise, difficulty in processing volume data and absence of computer tools. Regan *et al.* (2012) described that tool issues, lack of guidance, complexity and political issues are several aspects that hinder a company to adopt a traceability system. Dediu *et al.* (2016) also argued

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that difficulty in changing the work process is the main constraint for a firm to adopt a traceability system. Nevertheless, a traceability system is considered successful if the company could overcome the ethical problem, as well as challenges in management such as consistency, standardization, domination of general affairs and training (Morana, 2016).

2.2 Halal traceability system

Halal traceability system emphasizes halal transparency as one of the additional points in which does not reflect in the generic definition of traceability system (Khan *et al.*, 2018). Halal traceability system needs to be conducted to increase the transparency in the production chain, through the involvement of producers, manufacturers and distributors whose responsibility is to ensure a product follows the standards of food safety and Shariah guidelines, which entails traceability system as an important tool to achieve all the standards (Ab Rashid and Bojei, 2019; Zulfakar *et al.*, 2014). Besides, being acted as a communication tool, a traceability system could be easily accessed and retrieved information about halal food and any items throughout the food supply chain (Suhaiza *et al.*, 2010). Rashid *et al.* (2018) described that the adoption of a halal traceability system is crucial to enhance the integrity of the halal food supply chain and the performance of large companies and SMEs. To enhance the performance, some of these companies prefer to go through certain processes before intending to adopt a traceability system (Hew *et al.*, 2020; Othman *et al.*, 2016). Besides, the availability of a traceability system in SMEs' business operations is important to connect perceived benefits and readiness for HAS, in complying with halal assurance (Othman *et al.*, 2016; Abd Rahman *et al.*, 2017).

2.3 Technology-organization-environment framework

Technology-organization-environment (TOE) framework is an organizational-level theory that explains how the firm aspect affects innovation adoption (Baker, 2012). The TOE framework comprises three different factors in the firm aspect that influence the intention to adopt, namely, TA, OA and EA (Baker, 2012; Duan *et al.*, 2017; Tornatzky *et al.*, 1990; Zhu *et al.*, 2002). The TA covers both technologies that are used by the firm or yet to be used by the firm, where all technologies are applicable for a firm (Baker, 2012; Duan *et al.*, 2017; Zhu *et al.*, 2002). The existing technologies in a firm are crucial as they could trigger the technology innovation of the firm (Baker, 2012; Collins *et al.*, 1988). A firm needs to cautiously contemplate the adoption of new technology innovation by considering the industry and market as it could either lead to a small or huge impact on the firm (Baker, 2012; Hamill, 1997; Paul, 1996).

The OA entails the characteristics and resources of the firm which encompasses relationship among employees, communication process in the firm and the number of frailty resources which, in some way influenced technology adoption (Baker, 2012; Duan *et al.*, 2017; Morgan and Finnegan, 2007; Zhu *et al.*, 2002). The size of the firm, human resource quality and the complexity of managerial structure, as well as the way it was centralized and formalized describe the definition of OA (Zhu *et al.*, 2002). The EA comprises of industrial structure, service providers of technology existence and government regulation (Baker, 2012; Duan *et al.*, 2017; Zhu *et al.*, 2002). Both competitors and the complexity to obtain resources also constitute the EA (Zhu *et al.*, 2002). Several factors that pressure a firm to adopt technologies include a business partner and competitors (Bouchard, 1993; Hart and Saunders, 1998; Kuan and Chau, 2001; Premkumar and Ramamurthy, 1995).

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2.4 Technology acceptance model

Technology acceptance model (TAM) had been modified from theory reasoned action (TRA) by Fred Davis in 1986, specifically to model the acceptance of information systems or technologies among users (Bach *et al.*, 2016; Lai, 2017; Pinho and Soares, 2011; Sivo *et al.*, 2018). The core of TAM is to determine the behavioral intention to adopt information systems (Bach *et al.*, 2016). The model encompasses two specific beliefs, namely, PU and PEOU (Chen *et al.*, 2017; Lai, 2017). The PU is referred to as the subjective probability of the potential users where the use of a system would improve their action (Davis, 1989). The PU alludes to the extent of users' insight that utilization of information technology would contribute a remarkable value for them (Ajzen, 1991; Amin *et al.*, 2014; Rouibah *et al.*, 2011). Besides, PU is elucidated as the degree of the users' beliefs where their job performance would be enhanced by using a system as it also positively affects the users' intention to adopt the system (Ajzen, 1991; Amin *et al.*, 2014; Chen *et al.*, 2007).

The PEOU, on the other hand, is referred to as the anticipation of users about a system to be effortless (Bach *et al.*, 2016; Lai, 2017). The PEOU is alluding to the perception of users that adopting technologies would necessitate their mental effort (Ajzen and Fishbein, 1980; Amin *et al.*, 2014; Rouibah *et al.*, 2011). That is, users intend to adopt new technology as they perceived it to be easy to use (Amin *et al.*, 2014; Morosan, 2012). Both PU and PEOU are the crucial variables in TAM (Chang *et al.*, 2017; Chen *et al.*, 2013) where both beliefs influence inceptive user's intention to use certain technologies (Amin *et al.*, 2014; Davis, 1989; Davis and Venkatesh, 1996; Detlor *et al.*, 2013; Hess *et al.*, 2014; Martins *et al.*, 2014; Morosan, 2012). External variables in TAM represent other factors that might affect PU and PEOU toward a system (Chang *et al.*, 2017; Chen *et al.*, 2017; Davis, 1989). In this study, the TA, OA and EA represent external variables in TAM. These external variables were tested against the variables of PU and PEOU.

2.5 Integration of technology-organization-environment framework and technology acceptance model

The integration of the TOE framework and TAM is essential to determine the intention of adopting information technology that showed a good fit between both models. This was in line with other studies that integrating the adoption of innovation with a framework or model was pertinent to a study (Das and Das, 2012; Hameed *et al.*, 2012; Teo *et al.*, 2009). Awa *et al.* (2011) suggested enhancing TAM by adding variables from TOE to describe and identify the innovation that would be adopted by the organizations. The authors claimed that the integration of both models constitutes a more comprehensive model that provides a better understanding in the anticipation of information technology adoption by users. On the other hand, a model to determine the innovative adoption process of information system security was proposed by Hameed and Arachchilage (2017) that combines diffusion of innovation theory (DOI), TAM, theory planned behavior (TPB) and TOE framework. According to the authors, the understanding of the information system security adoption among organizations was improved by the theories and the integration of the models. It shows that integration of the theories and the models could enhance and contribute better results for the anticipation of innovation adoption among the users (Ashraf *et al.*, 2014; Awa *et al.*, 2011; Hameed and Arachchilage, 2017; Hameed *et al.*, 2012; Teo *et al.*, 2009).

Based on the discussions above on the model and framework, the following hypotheses were proposed:

H1a-e. There are associations between firmographic profiles and the intention of food SMEs to adopt a halal traceability system.

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H2a. There is a relationship between TA and PU.

H2b. There is a relationship between OA and PU.

H2c. There is a relationship between EA and PU.

H3a. There is a relationship between TA and PEOU.

H3b. There is a relationship between OA and PEOU.

H3c. There is a relationship between EA and PEOU.

3. Methodology

The TAM is widely used to identify the users' acceptance of an information system (Chen *et al.*, 2017; Lai, 2017). On the other hand, the TOE framework is an organizational-level theory that describes the adoption of an innovation that is affected by the firm aspect (Baker, 2012; Chen, 2019). A better result could be enhanced in the anticipation of innovation adoption among the users by integrating the theories and the models of the relevant topic (Awa *et al.*, 2011; Hameed and Arachchilage, 2017). Hence, TAM and TOE were integrated for this study to determine the intention of food SMEs to adopt a halal traceability system. The dependent variable for this study was the intention of food SMEs to adopt a halal traceability system while the independent variables were external factors which comprised of TA, OA, EA, PU and PEOU. The conceptual framework established for this study is illustrated in Figure 1.

Data collection was conducted throughout Peninsular Malaysia, involving 260 halal-certified food SMEs. The respondents were selected using a systematic random sampling method based on the data provided by the Department of Islamic Development Malaysia (JAKIM). The random choice was done which means randomly selecting a company between 1st and 20th of companies that manufactured food products, beverages and supplements for every 20 companies along with the list provided by JAKIM. Any number of ways could be done in selecting the first element of the sample as the particular sample is not considered as random because at the moment the first element was selected, the

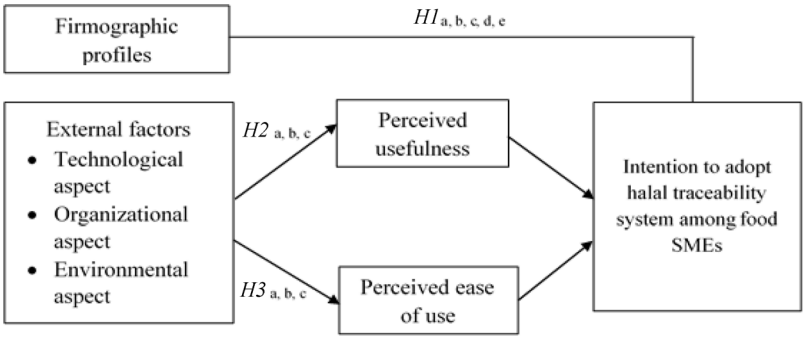

Figure 1. Conceptual framework of intention to adopt a halal traceability system among food SMEs

Sources: Tornatzky *et al.* (1990); Venkatesh and Davis (1996); Radipere and Dhliwayo (2014); Das and Das (2012); Anderson and Bolton (2015); Morck *et al.* (2005)

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remaining elements in the population would have either a likelihood of being chosen or not being chosen (Stopher, 2012). The research instrument was developed according to the framework constructed in which comprises the questions of firmographic profiles, TA, OA, EA, PU of a halal traceability system, PEOU of a halal traceability system and intention to adopt a halal traceability system. Options and open-ended questions were used to determine firmographic profiles, meanwhile, the closed-ended statements were established based on a five-point Likert Scale, ranging from 1 – strongly disagree to 5 – strongly agree. All the statements established to measure TA, OA, EA, PU of a halal traceability system, PEOU of halal traceability system and intention to adopt a halal traceability system were adapted from the past research studies as depicted in Table 1.

The data were analyzed using several statistical analyzes, namely, descriptive analysis, Chi-square analysis, Pearson correlation analysis and logistic regression analysis. Descriptive analysis was widely used by researchers (Ahmad Tarmizi *et al.*, 2020; Muhamad *et al.*, 2020; Lorente-Martínez *et al.*, 2020; Dincă *et al.*, 2019; Bernhard *et al.*, 2019; Raharja *et al.*, 2019; Agwu, 2018) to analyze the frequency of data to characterize the firmographic profiles of food SMEs and their level of intention to adopt a halal traceability system. Chi-square analysis, which was also used by other studies, to determine the association between firmographic profiles of food SMEs and their intention to adopt a halal traceability system (Kusuma *et al.*, 2020; Nguyen and Luu, 2020; Ali and Yusuf, 2019; Al Buraiki and Khan, 2018; Khan *et al.*, 2017; Ntwoku, 2017; Ali, 2016). The relationship between TA, OA and EA toward PU and PEOU was determined by Pearson correlation analysis. Pearson correlation analysis was commonly used (Ahmad Tarmizi *et al.*, 2020; Roberts-Lombard, 2020; Bilal *et al.*, 2020; Bananuka *et al.*, 2019; Jameel and Ahmad, 2018; Suki and Salleh, 2018; Janahi and Al Mubarak, 2017) to determine and measure the strength of the relationship between the variables tested. Meanwhile, logistic regression analysis was frequently used (Aghaei and Sokhanvar, 2020; AL-Shboul, 2019; Dincă *et al.*, 2019; Radicic and Djalilov, 2019; Appiah *et al.*, 2018; Awa *et al.*, 2017; Giotopoulos *et al.*, 2017) to anticipate the categorical outcomes. In an instance where the dependent variable is categorical, i.e. 0 and 1, logistic regression analysis is deemed as appropriate. It is also beneficial whenever some or all independent variables are either dichotomous or continuous (Pallant, 2005; Leech *et al.*, 2015). In this study, logistic regression analysis was used to investigate the most influential factors of food SMEs' intention to adopt a halal traceability system. A model was developed to investigate the factors that influence food SMEs' intention to adopt a halal traceability system.

4. Results and discussions

4.1 Firmographic profiles

Basic characteristics of the data for firmographic profiles of food SMEs were analyzed using descriptive analysis to illustrate the size of the company, legal status, year of establishment, sales turnover and region (Table 2). The results revealed that out of 260 food SMEs that were involved in this study, 207 companies (79.6%) were categorized as small size, 43 companies (16.5%) were the micro size and the remaining 10 companies (3.8%) were categorized as medium size. The size was classified according to the SME's definition, which is based on the number of employees (SME Corporation Malaysia, 2018). About 113 companies (43.5%) were categorized as sole proprietors, whereas 118 companies (45.4%) and 29 companies (11.2%) were categorized as private limited and partnership, respectively. Most of the food SMEs were established for less or equal to 9 years (n = 150, 57.7%), whereas the remaining food SMEs were established for more than 9 years (n = 110, 42.3%). The sales turnover, which was also categorized according to SME definition, indicated that 142 food SMEs (54.6%) have gained less than RM300,000 per year, 115 (44.2%) have gained between RM300,000 to RM14.9m and

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Variable	Statement	Source
Technological aspect (TA)	1. We have the necessary skills and knowledge to use a halal traceability system	Thong and Yap (1995)
	2. We have the experience of using a system that similar to the halal traceability system	
	3. We are confident in using a halal traceability system even though we do not have experienced using it	
	4. It is easy for us to adopt a halal traceability system as we have experience in adopting various systems in managing our business	
	5. Our IT department employees have adequate skills for halal traceability system adoption	Bach *et al.* (2016)
	6. We have the skills in using a system that same as a halal traceability system	
	7. Adoption of a halal traceability system would improve the operational efficiency of our firm	Kuan and Chau (2001)
	8. We can improve the image of our firm if a halal traceability system is adopted	
Organizational aspect (OA)	1. Top management would support the implementation of a halal traceability system	Thong *et al.* (1996)
	2. Top management would allocate adequate resources for the adoption of a halal traceability system	
	3. Top management would encourage employees to use a halal traceability system	
	4. Our firm needs to incur high training costs to train our employees if we adopt a halal traceability system	Kuan and Chau (2001)
	5. It is easy to adopt a halal traceability system in our firm as we have sufficient expertise	
	6. The use of the halal traceability system could assist the firm to increase the profit margin	
	7. Our firm needs to bear high implementation costs if we implement a halal traceability system	
	8. The use of a halal traceability system would assist our firm to reduce the operating cost	
Environmental aspect (EA)	1. Government supports halal traceability system implementation	Ramsey and McCole (2005)
	2. Government provides incentives for SMEs to implement new technology as such a halal traceability system	
	3. Government provides information and encourages us to implement technology such as a halal traceability system in managing the business	
	4. We will gain more customers compared to competitors if we implement a halal traceability system	Thong (1999); Al-Qirim and Corbitt (2002)
	5. Halal traceability system is a strategic necessity to compete in the marketplace	
	6. We will increase our market share if we implement a halal traceability system	
	7. Government advises SMEs to implement a halal traceability system	Kuan and Chau (2001)
	8. Our competitors using a system that similar to the halal traceability system in managing their business	

(*continued*)

Table 1. Statements on technological aspect, organizational aspect, environmental aspect, perceived usefulness, perceived ease of use and intention

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Variable	Statement	Source
Perceived usefulness (PU)	1. Using a halal traceability system improves company performance	Bach *et al.* (2016)
	2. Using a halal traceability system increase company productivity	
	3. We found that a halal traceability system is useful to our company	
	4. Using a halal traceability system improves employee performance	
	5. Using a halal traceability system would allow the firm to trace products along the food supply chain more quickly	Chang *et al.* (2017)
	6. Using a halal traceability system would enhance effectiveness in tracing product materials	
	7. The adoption of a halal traceability system will provide timely information required for decision-making	Premkumar and Roberts (1999)
	8. Halal traceability system is compatible with our organization's current values and goals	
Perceived ease of use (PEOU)	1. Implementation process of the halal traceability system is understandable	Bach *et al.* (2016)
	2. It is easy to integrate a halal traceability system with the existing system	
	3. We found that it is easy to learn about the halal traceability system	
	4. Management using a halal traceability system would be clearer and more understandable	Chang *et al.* (2017)
	5. The skills required to use a halal traceability system are simple for our employees	Premkumar and Roberts (1999)
	6. The complexity of the halal traceability system is low	
	7. Halal traceability system allows our firm to trace products along the food supply chain easily	Butt *et al.* (2016)
	8. Overall, the halal traceability system is easy to use	Amin *et al.* (2014)
Intention to adopt a halal traceability system	1. We have a high intention to use a halal traceability system in our firm	Elbeltagi *et al.* (2013)
	2. We intend to learn about using a halal traceability system	
	3. We intend to adopt a halal traceability system	
	4. We plan to use a halal traceability system to manage our business	
	5. We predict that we would use a halal traceability system in the future	Chang *et al.* (2017)
	6. We will take initiative to adopt a halal traceability system in the future	
	7. We intend to adopt a halal traceability system as the profit gains by the firm is expected to be increased	
	8. We would use a halal traceability system rather than the old method to trace the products	Agarwal and Prasad (1998)

Table 1.

only three (3.2%) of the food SMEs have gained between RM15m to RM50m. Central region (n = 111, 42.7%) led the most distributed region of food SMEs, followed by Northern region (n = 62, 23.8%), East Coast region (n = 52, 20.0%) and Southern region (n = 35, 13.5%).

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Table 2. Firmographic profiles of food SMEs

Profile	Frequency (*n*)	Percentage (%)
Size		
Medium	10	3.8
Small	207	79.6
Micro	43	16.5
Year of establishment		
≤9 years	150	57.7
>9 years	110	42.3
Sales turnover		
RM15m ≤ RM50m	3	1.2
RM300,000 < RM15m	115	44.2
< RM300,000	142	54.6
Business ownership		
Sole proprietor	113	43.5
Partnership	29	11.2
Limited company	118	45.4
Region		
Northern	62	23.8
East Coast	52	20.0
Central	111	42.7
Southern	35	13.5

4.2 Mean ranking analysis

Mean ranking analysis was used to measure and describe the independent and dependent variables, namely, TA, OA, EA, PU, PEOU and intention of food SMEs to adopt a halal traceability system. The results of these variables were then used as part of variables in other statistical analyzes such as Pearson correlation analysis and logistic regression analysis. All the statements for each variable were established based on 5-point Likert scales, namely, 1-strongly disagree; 2-disagree; 3-neutral; 4-agree; 5-strongly agree. The results of the statements in respective tables were ranked from the lowest to the highest mean score.

4.2.1 Technological aspect. Eight statements related to TA were analyzed and the results of the mean scores are summarized in Table 3. Based on Table 3, the highest mean score of 4.30 was for the statement "Adoption of a halal traceability system would improve the operational efficiency of our firm," while the lowest mean score of 4.08 was for the statement "We can improve the image of our firm if a halal traceability system is adopted." The average mean score was 4.18, indicating that the majority of the respondents have high agreement toward TA of a halal traceability system. This finding is supported by earlier studies suggesting that TA is one of the important elements of a traceability system (Thong and Yap, 1995; Bach *et al.*, 2016; Kuan and Chau, 2001).

4.2.2 Organizational aspect. Table 4 summarizes the mean scores of the eight statements related to OA. The statements were ranked from the highest to the lowest. The highest mean score of 4.28 was for the statement "Top management would allocate adequate resources for the adoption of a halal traceability system." While "The use of a halal traceability system would assist our firm to reduce the operating

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No.	Statement	Frequency (*n*)/percentage (%) 1	2	3	4	5	Mean	Standard deviation
1	Adoption of a halal traceability system would improve the operational efficiency of our firm	1(0.4)	6(2.3)	11(4.2)	139(53.5)	103(39.6)	4.30	0.692
2	We are confident in using a halal traceability system even though we do not have experienced using it	0(0.0)	8(3.1)	21(8.1)	126(48.5)	105(40.4)	4.26	0.736
3	We have the experience of using a system that similar to the halal traceability system	1(0.4)	9(3.5)	29(11.2)	118(45.4)	103(39.6)	4.20	0.801
4	We have the necessary skills and knowledge to use a halal traceability system	0(0.0)	6(2.3)	32(12.3)	136(52.3)	86(33.1)	4.16	0.723
5	We have the skills in using a system that same as a halal traceability system	1(0.4)	8(3.1)	23(8.8)	146(56.2)	82(31.5)	4.16	0.721
6	It is easy for us to adopt a halal traceability system as we have experience in adopting various systems in managing our business	3(1.2)	15(5.8)	22(8.5)	120(46.2)	100(38.5)	4.15	0.886
7	Our IT department employees have adequate skills for a halal traceability system adoption	17(6.5)	42(16.2)	39(15.0)	109(46.9)	53(20.4)	4.13	0.753
8	We can improve the image of our firm if a halal traceability system is adopted	0(0.0)	10(3.8)	37(14.2)	136(52.3)	77(29.6)	4.08	0.767
	Average mean and standard deviation						4.18	0.768

Table 3. Technological aspect

cost" was the statement with the lowest mean score of 4.10. The average mean score of 4.19, suggesting that the majority of the respondents indicated their high agreement on OA of halal traceability system and the finding is similar to previous studies of Thong *et al.* (1996); Kuan and Chau (2001) that OA should be a priority toward the success of traceability systems implementation.

4.2.3 Environmental aspect. The mean scores of the eight statements related to EA of halal traceability system are summarized in Table 5. The highest mean score of 4.26 was for the statement "Halal traceability system is a strategic necessity to compete in the marketplace." The lowest mean score of 4.04 was indicated by the statement "Our competitors using a system that similar to the halal traceability system in managing their business." The average mean score of the eight statements related to EA was 4.15, indicating that the majority of the respondents were agreed EA is important for implementing a halal traceability system. This was in line with previous studies by Ramsey and McCole (2005); Thong (1999); Al-Qirim and Corbitt (2002); Kuan and Chau (2001) that EA is an important element of a traceability system.

4.2.4 Perceived usefulness of a halal traceability system. Eight statements related to PU of halal traceability system were analyzed. Table 6 summarizes the mean scores of PU

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Table 4. Organizational aspect

No.	Statement	Frequency (*n*)/percentage (%) 1	2	3	4	5	Mean	Standard deviation
1	Top management would allocate adequate resources for the adoption of a halal traceability system	1(0.4)	6(2.3)	14(5.4)	137(52.7)	102(39.2)	4.28	0.704
2	Top management would support the implementation of a halal traceability system	2(0.8)	8(3.1)	21(8.1)	120(46.2)	109(41.9)	4.25	0.794
3	Top management would encourage employees to use a halal traceability system	0(0.0)	3(1.2)	14(5.4)	160(61.5)	83(31.9)	4.24	0.602
4	Our firm needs to incur high training costs to train our employees if we adopt a halal traceability system	3(1.2)	3(1.2)	24(9.2)	140(53.8)	90(34.6)	4.20	0.743
5	It is easy to adopt a halal traceability system in our firm as we have sufficient expertise	0(0.0)	5(1.9)	24(9.2)	150(57.7)	81(31.2)	4.18	0.671
6	The use of the halal traceability system could assist the firm to increase the profit margin	1(0.4)	3(1.2)	37(14.2)	129(49.6)	90(34.6)	4.17	0.723
7	Our firm needs to bear high implementation costs if we implement a halal traceability system	2(0.8)	8(3.1)	23(8.8)	149(57.3)	78(30.0)	4.13	0.753
8	The use of a halal traceability system would assist our firm to reduce the operating cost	3(1.2)	13(5.0)	35(13.5)	114(43.8)	95(36.5)	4.10	0.893
	Average mean and standard deviation						4.19	0.735

of a halal traceability system. "Using a halal traceability system would enhance effectiveness in tracing product materials" was the statement with the highest mean score of 4.25, while the lowest mean score of 4.05 was for the statement "Using halal traceability system improves employee performance." The average mean score of 4.19 for the eight statements showed a high agreement of the respondents toward the halal traceability system that was perceived to be useful. This finding is supported by earlier studies suggesting that a traceability system is important and perceived to be useful by firms that have successfully implemented the system (Bach *et al.*, 2016; Chang *et al.*, 2017; Premkumar and Roberts, 1999).

4.2.5 Perceived ease of use of halal traceability system. Table 7 Summarizes the mean scores of the eight statements related to PEOU of halal traceability system which was ranked from the highest to the lowest mean accordingly. The statements of "halal traceability system allow our firm to trace products along the food supply chain easily" showed the highest mean score of 4.36, while "the complexity of halal traceability system is low" was the lowest mean score of 4.10. The average mean score was 4.17, reflecting the respondents' agreement toward PEOU of a halal traceability system. This finding was in line with earlier studies of Bach *et al.* (2016), Chang *et al.* (2017); Premkumar and Roberts (1999); Butt *et al.* (2016); Amin *et al.* (2014) that the ease of use of traceability system has been described as one of the advantages of the system.

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No.	Statement	Frequency (*n*)/percentage (%) 1	2	3	4	5	Mean	Standard deviation
1	Halal traceability system is a strategic necessity to compete in the marketplace	0(0.0)	2(0.8)	20(7.7)	146(56.2)	92(35.4)	4.26	0.628
2	Government provides information and encourages us to implement technology such as a halal traceability system in managing the business	0(0.0)	5(1.9)	22(8.5)	146(56.2)	87(33.5)	4.21	0.673
3	Government supports halal traceability system implementation	2(0.8)	7(2.7)	22(8.5)	137(52.7)	92(35.4)	4.19	0.762
4	Government provides incentives for SMEs to implement new technology as such a halal traceability system	0(0.0)	9(3.5)	23(8.8)	146(56.2)	82(31.5)	4.16	0.721
5	We will gain more customers compared to competitors if we implement a halal traceability system	5(1.9)	11(4.2)	22(8.5)	130(50.0)	92(35.4)	4.13	0.876
6	We will increase our market share if we implement a halal traceability system	5(1.9)	8(3.1)	23(8.8)	141(54.2)	83(31.9)	4.11	0.833
7	Government advises SMEs to implement a halal traceability system	0(0.0)	9(3.5)	21(8.1)	172(66.2)	58(22.3)	4.07	0.662
8	Our competitors using a system that similar to the halal traceability system in managing their business	3(1.2)	10(3.8)	26(10.0)	155(59.6)	66(25.4)	4.04	0.782
	Average mean and standard deviation						4.15	0.742

Table 5. Environmental aspect

No.	Statement	Frequency (*n*)/percentage (%) 1	2	3	4	5	Mean	Standard deviation
1	Using a halal traceability system would enhance effectiveness in tracing product materials	1(0.4)	1(0.4)	33(12.7)	122(46.9)	103(39.6)	4.25	0.716
2	Using a halal traceability system increase company productivity	2(0.8)	6(2.3)	27(10.4)	120(46.2)	105(40.4)	4.23	0.786
3	Halal traceability system is compatible with our firm's current values and goals	2(0.8)	5(1.9)	30(11.5)	117(45.0)	106(40.8)	4.23	0.786
4	Using a halal traceability system would allow the firm to trace products along the food supply chain more quickly	1(0.4)	4(1.5)	35(13.5)	122(46.9)	98(37.7)	4.20	0.755
5	Using a halal traceability system improves company performance	1(0.4)	7(2.7)	27(10.4)	130(50.0)	95(36.5)	4.20	0.758
6	We found that a halal traceability system is useful to our company	1(0.4)	4(1.5)	29(11.2)	136(52.3)	90(34.6)	4.19	0.720
7	The adoption of a halal traceability system will provide timely information required for decision-making	2(0.8)	2(0.8)	45(17.3)	122(46.9)	89(34.2)	4.13	0.775
8	Using a halal traceability system improves employee performance	2(0.8)	7(2.7)	50(19.2)	119(45.8)	82(31.5)	4.05	0.828
	Average mean and standard deviation						4.19	0.766

Notes: 1-strongly disagree; 2-disagree; 3-neutral; 4-agree; 5-strongly agree

Table 6. Perceived usefulness of a halal traceability system

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Table 7. Perceived ease of use of a halal traceability system

No.	Statements	Frequency (*n*)/percentage (%) 1	2	3	4	5	Mean	Standard deviation
1	Halal traceability system allows our firm to trace products along the food supply chain easily	1(0.4)	3(1.2)	51(19.6)	129(49.6)	76(29.2)	4.36	0.753
2	Management using a halal traceability system would be clearer and more understandable	2(0.8)	7(2.7)	47(18.1)	129(49.6)	75(28.8)	4.23	0.805
3	Overall, a halal traceability system is easy to use	3(1.2)	10(3.8)	56(21.5)	115(44.2)	76(29.2)	4.17	0.876
4	Implementation process of a halal traceability system is understandable	3(1.2)	7(2.7)	52(20.0)	135(51.9)	63(24.2)	4.15	0.809
5	The skills required to use a halal traceability system are simple for our employees	5(1.9)	9(3.5)	52(20.0)	122(46.9)	72(27.7)	4.15	0.888
6	It is easy to integrate a halal traceability system with existing system	4(1.5)	7(2.7)	60(23.1)	123(47.3)	66(25.4)	4.12	0.853
7	We found that it is easy to learn about a halal traceability system	4(1.5)	10(3.8)	56(21.5)	127(48.8)	63(24.2)	4.11	0.862
8	The complexity of a halal traceability system is low	7(2.7)	18(6.9)	75(28.8)	104(40.0)	56(21.5)	4.10	0.970
	Average mean and standard deviation						4.17	0.820

Notes: 1-strongly disagree; 2-disagree; 3-neutral; 4-agree; 5-strongly agree

Table 8. Food SMEs' intention to adopt a halal traceability system

No.	Statements	Frequency (*n*)/percentage (%) 1	2	3	4	5	Mean	Standard deviation
1	We intend to adopt a halal traceability system	3(1.2)	4(1.5)	36(13.8)	113(43.5)	104(40.0)	4.20	0.817
2	We intend to learn about using a halal traceability system	3(1.2)	4(1.5)	30(11.5)	127(48.8)	96(36.9)	4.19	0.785
3	We predict that we would use a halal traceability system in the future	4(1.5)	4(1.5)	40(15.4)	104(40.0)	108(41.5)	4.18	0.859
4	We plan to use a halal traceability system to manage our business	2(0.8)	5(1.9)	39(15.0)	119(45.8)	95(36.5)	4.15	0.800
5	We will take initiative to adopt a halal traceability system in the future	5(1.9)	1(0.4)	42(16.2)	118(45.4)	94(36.2)	4.13	0.834
6	We intend to adopt a halal traceability system as the profit gains by the firm are expected to increase	4(1.5)	0(0.00)	49(18.8)	111(42.7)	96(36.9)	4.13	0.825
7	We would use a halal traceability system rather than the old method to trace the products	3(1.2)	4(1.5)	40(15.4)	127(48.8)	86(33.1)	4.11	0.800
8	We have a high intention to use a halal traceability system in our firm	3(1.2)	7(2.7)	43(16.5)	121(46.5)	86(33.1)	4.08	0.839
	Average mean and standard deviation						4.15	0.820

Notes: 1-strongly disagree; 2-disagree; 3-neutral; 4-agree; 5-strongly agree

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4.2.6 Intention to adopt a halal traceability system. Table 8 summarizes the mean scores of the eight statements related to the intention of food SMEs to adopt a halal traceability system, in which the statements were ranked from the highest to the lowest mean, respectively. The highest mean score of 4.20 was for the statement "We intend to adopt a halal traceability system." While "We have a high intention to use a halal traceability system in our firm" was the statement with the lowest mean score of 4.08. The average mean score for the eight statements was 4.15, suggesting that the respondents have a high intention to adopt the system. As indicated in relevant studies such as Elbeltagi *et al.* (2013); Chang *et al.* (2017); Agarwal and Prasad (1998) that firms have a high interest in adopting advanced systems to assist and manage their business.

In this study, the scale was categorized into high and low intention where the mean score of 4.00 and above was considered as a high level of intention whereas the mean score below 4.00 was considered as a low level of intention. This is in line with the intention scales suggested by López-Núñez *et al.* (2020), Iacobucci *et al.* (2015a, 2015b) and Carmeli (2001). The result of the intention level to adopt a halal traceability system is summarized in Table 9.

The majority of food SMEs (n = 215, 82.7%) have a high intention to adopt a halal traceability system in performing their business while the remaining food SMEs (n = 45, 17.3%) have a low intention to adopt a halal traceability system to perform their business. This result was indirectly contributed by PU and PEOU of the halal traceability system. As described by Taherdoost (2018), PU and PEOU showed a notable impact on the intention. Hence, PU and PEOU were proven to influence the intention of inceptive users to use certain technologies (Amin *et al.*, 2014; Davis, 1989; Detlor *et al.*, 2013; Hess *et al.*, 2014; Martins *et al.*, 2014; Morosan, 2012; Naqvi *et al.*, 2019; Upadhyay *et al.*, 2018). That is, the development of positive intention to adopt a halal traceability system depends on both the PU and PEOU of food SMEs toward halal traceability systems.

4.5 Association between firmographic profiles and food small and medium enterprises' intention to adopt a halal traceability system

Based on Chi-square analysis, there was a significant association between sales turnover and food SMEs' intention to adopt a halal traceability system at a 1% level of significance

Intention level (mean score)	Frequency (n)	Percentage (%)
Low intention (<4)	45	17.3
High intention (≥4)	215	82.7

Table 9. Intention level of food SMEs to adopt a halal traceability system

Variables	Chi-square value	df	Significance	Decision
Size	0.517	2	0.772	Fail to reject H_0
Year of establishment	0.507	1	0.486	Fail to reject H_0
Sales turnover	13.113	2	0.001***	Reject H_0
Business ownership	0.904	2	0.636	Fail to reject H_0
Region	7.654	3	0.054**	Reject H_0

Notes: ***Significant at 1% level of significance; **significant at 5% level of significance

Table 10. Chi-square analysis

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(Table 10). Thus, we reject the null hypothesis (H_0), indicating that there was a significant association between sales turnover and the intention of food SMEs to adopt a halal traceability system. This describes that sales turnover plays a significant role in ensuring the adoption of a halal traceability system among food SMEs. This finding was in line with Ahmad Tarmizi *et al.* (2014), Hassan *et al.* (2017) and Mohamad *et al.* (2020) who claimed that SMEs that possessed additional resources projected for information technology (IT) tend to adopt new technology. A significant association was also shown between the region and food SMEs' intention to adopt a halal traceability system at a 1% level of significance Thus, the null hypothesis (H_0) was rejected, indicating that there was a significant association between region and the intention of food SMEs to adopt a halal traceability system. The adoption of technology is imperative throughout the various location to increase knowledge flows within geographically different organizations (Forman and Zeebroeck, 2019). Thus, the adoption of a halal traceability system, which is one of the important technologies needed in a firm, could promote knowledge flows within geographically different organizations. These associations were significant at 1% and 5% levels of significance, respectively.

4.6 Relationship between technological aspect, organizational aspect and environmental aspect toward perceived usefulness and perceived ease of use

Pearson correlation analysis was used to determine the relationship and strength between TA, OA and EA toward PU and PEOU. The results revealed significant values of all variables tested that affirm the existence of the relationship between TA, OA and EA toward PU and PEOU. The strength of the relationship between the variables is indicated by the r value as illustrated in Table 11. The rules of thumb for correlation coefficient size are suggested by Hair *et al.* (2007). Based on the Pearson correlation value stated in Table 11, shows that there was a strong correlation between EA and PU ($r = 0.720$, $p < 0.01$). The result described that the EA, which covered government and competitors in this study, has a crucial role to ensure that the usefulness of the halal traceability system was perceived by food SMEs. This is consistent with the argument made by Maynard (2007) that the government should consider expanding the economic gains beyond capital while ascertaining the transfer of technology and expertise from MNCs to local firms. Besides, SMEs confronting pressure from competitors tend to adopt the technology (Hassan *et al.*, 2017). Meanwhile, the other relationships showed moderate correlations, namely, between OA and PU ($r = 0.693$, $p < 0.01$); TA and PEOU ($r = 0.688$, $p < 0.01$); OA and PEOU ($r = 0.687$, $p < 0.01$); EA and PEOU ($r = 0.641$, $p < 0.01$) and TA and PU ($r = 0.556$, $p < 0.01$). Based on the results, the null hypothesis (H_0) was rejected, indicating that there were relationships between TA, OA, EA, PU and PEOU. All the relationships were statistically significant at a 1% level of significance.

Variables		Technological aspect (TA)	Organizational aspect (OA)	Environmental aspect (EA)
Perceived usefulness (PU)	Pearson correlation	0.556	0.693	0.720
	Significance	0.000***	0.000***	0.000***
	Decision	Reject H_0	Reject H_0	Reject H_0
Perceived ease of use (PEOU)	Pearson correlation	0.688	0.687	0.641
	Significance	0.000***	0.000***	0.000***
	Decision	Reject H_0	Reject H_0	Reject H_0

Note: ***Correlation is significant at the 0.01 level (two-tailed)

Table 11.
Correlation analysis

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4.7 Factors influencing food small and medium enterprises' intention to adopt a halal traceability system

Logistic regression analysis was used to identify the most influencing factors of food SMEs' intention to adopt a halal traceability system. The independent variables tested were firmographic profiles (sales turnover, region), PU, PEOU, TA, OA and EA while food SMEs' intention to adopt a halal traceability system was tested as the dependent variable. Table 12 shows the coding system for the measured variables for this study. The dependent variable, food SMEs' intention to adopt a halal traceability system was categorized into "0-low intention" and "1-high intention." As the dependent variable tested was required to be on a dichotomous scale, the mean score of intention was divided into two levels of intention, namely, low and high.

Meanwhile, seven independent variables were tested to determine the most influential factors on food SMEs' intention to adopt a halal traceability system. The predictors, which are independent variables, consisted of categorical and continuous variables. The categorical variables i.e. firmographic profiles were categorized into 1 and 0, specifically, sales turnover (small size = 1, otherwise = 0); and region (Central region = 1, otherwise = 0). On the other hand, the continuous variables, namely, PU, PEOU, TA, OA and EA were tested using the mean score.

Based on the result of the Hosmer and Lemeshow Test, the Chi-square and significant values were 15.733 and 0.185, respectively, where the significant value exceeds 0.05. This indicates the goodness of fit of the data or in other words, the model fitted the data well. The indication of the amount of variation in the dependent variable explained can be obtained from the Cox and Snell R Square and the Nagelkerke R Square values. The results in Table 13 further showed that the values of the Cox and Snell R Square and the Nagelkerke R Square were 0.383 and 0.585, respectively. Meanwhile, the values that explained the percentage of variability by this set of variables ranged between 38.3% and 58.5%. The Wald value shows how important each of the significant factors affects food SMEs' intention to adopt a halal traceability system. The highest Wald value was 26.474, indicating that PU showed the most effective variable while the region has the least effect variable with a Wald value of 1.049.

Based on Table 13, six out of the seven variables tested have significant values, namely, PU, PEOU, TA, OA, EA and sales turnover, at 1%, 5% and 10% levels of significance, respectively. The probability for a case to fall into a specific category was explained by the exponential beta value (Pallant, 2005). Meanwhile, the positive and negative signs of the

Variables		Coding System
Dependent variable	Intention to adopt a halal traceability system	1 = high intention 0 = low intention
Independent variables	Sales turnover	1 = small size 0 = otherwise
	Region	1 = central region 0 = otherwise
	Perceived usefulness (PU)	Dichotomous (mean score)
	Perceived ease of use (PEOU)	Dichotomous (mean score)
	Technological aspect (TA)	Dichotomous (mean score)
	Organizational aspect (OA)	Dichotomous (mean score)
	Environmental aspect (EA)	Dichotomous (mean score)

Table 12. Coding system for food SMEs' intention to adopt a halal traceability system

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Table 13. The summary result of logistic regression analysis

Variables	Estimated coefficient	Wald	Significant level	Exp (B)
Region	0.842	1.049	0.825	1.138
Sales turnover	0.901	5.531	0.018**	2.462
Perceived usefulness (PU)	2.583	26.474	0.000***	13.242
Perceived ease of use (PEOU)	1.253	9.053	0.003***	3.500
Technological aspect (TA)	0.928	13.212	0.001***	2.555
Organizational aspect (OA)	1.395	2.623	0.056*	2.043
Environmental aspect (EA)	0.623	6.786	0.067*	1.528
Constant	8.733	48.270	0.000	0.000

Model Summary			
−2 log-likelihood	382.481	Hosmer and Lemeshow test	15.733
Cox and Snell *R* square	0.383	df	8
Nagelkerke *R* square	0.585	Sig.	0.185

Notes: ***Significant at 1% level of significance; **significant at 5% level of significance; *significant at 1% level of significance; DV: low intention/high intention

values at the estimated coefficient explained the relationship direction within the factors. All the directions of the relationships were positive. Based on the results in Table 13, food SMEs were 13.242 times intend to adopt a halal traceability system if the system is perceived to be useful. The result proved that PU's argument represents the extent of users' prediction that the use of information technology will contribute exceptional benefits to them (Ajzen, 1991; Amin *et al.*, 2014; Rouibah *et al.*, 2011).

The results further revealed that food SMEs were 3.500 times intend to adopt a halal traceability system if the handling of the system is perceived to be easy. The result was parallel with a premise that claimed users would intend to adopt new technology if they perceived the technology is easy to be used (Amin *et al.*, 2014; Morosan, 2012). Another variable that has a significant value was TA, indicating that food SMEs were 2.555 times intend to adopt a halal traceability system due to TA of the system. As argued by Bach *et al.* (2016), SMEs will show their interest in adopting traceability systems due to their TA that could improve business competitiveness. Food SMEs' intention to adopt a halal traceability system will also be increased by 2.462 times if the sales turnover of the companies is increased. According to Hassan *et al.* (2017), SMEs with supplementary resources allocated for information technology (IT), have a high tendency to adopt new technology. OA is deemed an important factor in managing SMEs. Based on the results, it showed that food SMEs have 2.043 times of intention to adopt a halal traceability system because of their understanding of the importance of OA. This finding is supported by the previous study of Kuan and Chau (2001) that it is easy to adopt a halal traceability system by SMEs that have sufficient expertise, which is one of the elements of OA. Finally, the results revealed that food SMEs were 1.528 times intend to adopt a halal traceability system in fulfilling EA, which is a strategic necessity to compete in the marketplace (Ramsey and McCole, 2005).

Meanwhile, one of the variables tested, namely, region, has no significant value. It indicates that region was not the influential factor on food SMEs' intention to adopt a halal traceability system. The following equation model summarized and showed that food SMEs' intention to adopt a halal traceability system was explained by PU, PEOU, TA, OA, EA and sales turnover:

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$$
\begin{aligned}
&\text{Y (food SMEs' intention to adopt a halal traceability system)} \\
&= 8.733 + 0.901\,(\text{sales turnover}) + 2.583\,(\text{perceived usefulness}) \\
&\quad + 1.253\,(\text{perceived ease of use}) + 0.928\,(\text{technological aspect}) \\
&\quad + 1.395\,(\text{organizational aspect}) + 0.623\,(\text{environmental aspect}) + \varepsilon
\end{aligned}
$$

5. Implications, limitations and conclusions

This study provides several important practical and theoretical insights. We found that food SMEs have a high intention to adopt a halal traceability system, indicating that they are open to any innovation that could enhance their business performance. Food SMEs is one of the major sectors that contribute to Malaysia's economic enhancement. Therefore, the government has provided a platform that promotes global recognition for SMEs such as SMEs Go Global Programme, to facilitate their expansion into international markets. The encouragement demonstrated by the government to push for food SMEs to be globally competitive needs to be aligned with the adoption of technological advances. The adoption is crucial to improve the food industry performance (Khairuddin *et al.*, 2020), especially for halal products. The violation of food halalness had raised concerns among Muslim consumers that composed the majority of the population in Malaysia. The advanced technology, particularly the halal traceability system could enhance the transparency of food products in the Malaysian halal market. The adoption of advanced technology could, therefore, provide the tremendous potential to further improve the halal industry in Malaysia. As halal food products undergo the entire verified supply chain, Muslim consumers would be more assured in their selection of dietary consumption, thereby implicitly reduce their concern regarding any violation of food halalness. As the global halal market continues to grow, the adoption of the halal traceability system by the Malaysian food SMEs would provide huge opportunities for their market expansion.

Food SMEs' performance could be improved through their active involvement and participation in several initiatives, as well as financial assistance provided by relevant agencies like SME Corporation Malaysia. Most importantly, the adoption of advanced technology is crucial in ensuring the knowledge flows within geographically different organizations where the food SMEs mostly located for easy accessibility and convenience. On top of that, EA has a significant role in providing a good insight into the usefulness of a halal traceability system. In the aspect of competitors, the government needs to increase the number of events for food SMEs to participate in. Market intelligence data that were gained from the events could help food SMEs to come out with accurate decision-making in identifying the relevant strategies to compete with their competitors. This study also revealed that PU, PEOU, TA, OA, EA and sales turnover influence food SMEs' intention to adopt a halal traceability system. As the government plays an essential role in providing adequate knowledge about halal traceability systems by increasing and intensifying the course about the system, food SMEs also need to be exposed to training on halal traceability systems and further encourage them to adopt a halal traceability system.

As the only organization responsible for halal, JAKIM needs to reinforce the effectiveness of adopting a halal traceability system and also guiding food SMEs to implement HAS in its business operation instead of merely encouraging. This action could expose and drive all categories of food SMEs with low intention to adopt a halal traceability system. Furthermore, SME Corporation Malaysia, which is an authority responsible for the SMEs' development, had introduced numerous initiatives that aim to assist SMEs in improving their business performance, parallel with their mission to promote the development of

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innovative, competitive, and resilient SMEs through effective provision and coordination of business support. Therefore, making these initiatives available for SMEs, indeed requires active involvement and participation of food SMEs for more great opportunities, which will lead to more competitive intensity.

This study reveals some interesting results. Nevertheless, it has several limitations that have to be considered. First, some of the food SMEs were reluctant to participate in this study due to the lack of experience and inadequate knowledge about the halal traceability system. Second, the sample size to represent food SMEs were deemed to be small and could not fully represent the population of food SMEs, thereby limiting the generalization of the findings. Hence, future research should increase the sample size to increase the representation of food SMEs in Malaysia to allow generalization of the findings. Also, the expansion of the location of the study to Sabah and Sarawak would enable the generalization of the Malaysian SME context. Furthermore, other variables need to be added for external factors and should be analyzed using structural equation modeling, which is a more advanced statistical analysis to identify and confirm more factors, for instance, knowledge, attitude and practices that may influence food SMEs' intention to adopt a halal traceability system. The integration of other theories such as DOI and TPB is also suggested for future research as the integration of these theories could increase the understanding of food SMEs toward halal traceability systems.

In a nutshell, the adoption of advanced technology by SMEs is crucial to improve the food and beverages industry performance, especially for halal products. Technology advances such as Blockchain, Industrial Revolution (IR) 4.0, Big Data, Internet of Things (IoT), Financial Technology (FinTech) and web-based halal traceability system could enhance the transparency of food products in the Malaysian halal market. These advanced technologies adoption could, therefore, provide a huge potential to further improve the halal industry in Malaysia. As halal food products undergo a verified supply chain entirely, Muslim consumers tend to be more assured in choosing their dietary consumption which could implicitly reduce their concern regarding any violation of food halalness.

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